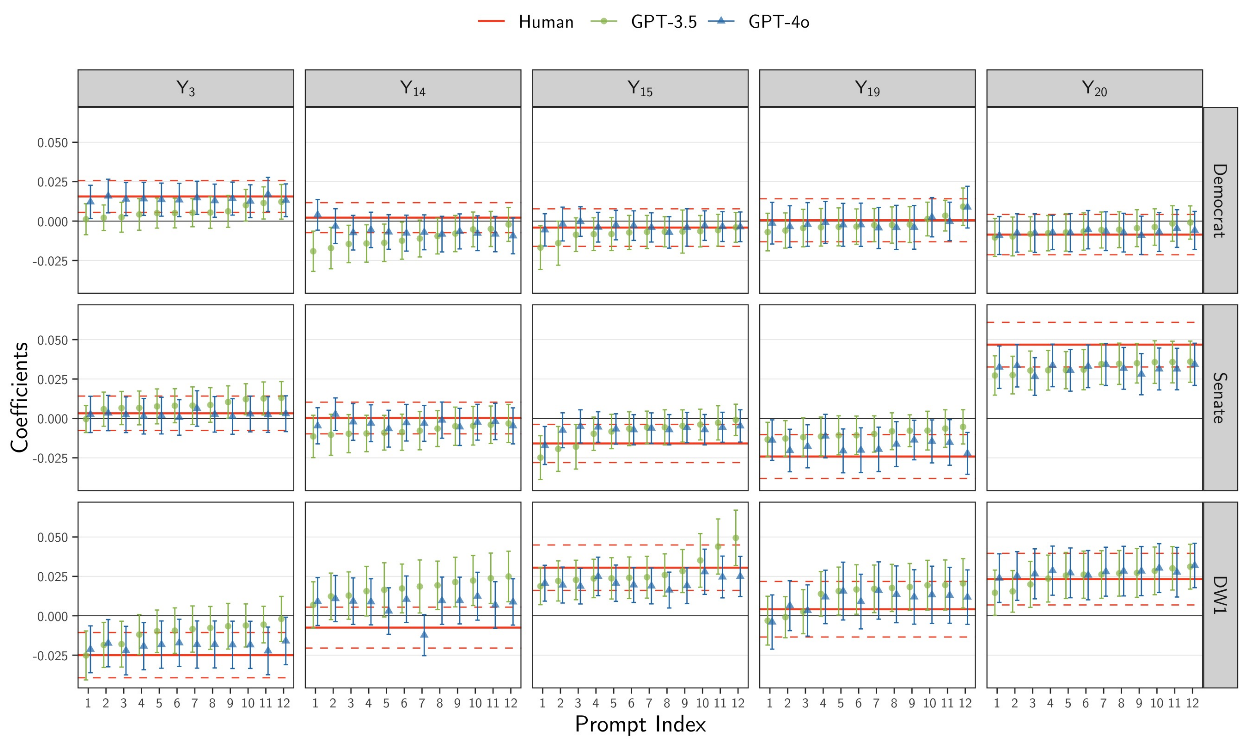
Results LHS

Aug 12, 2024

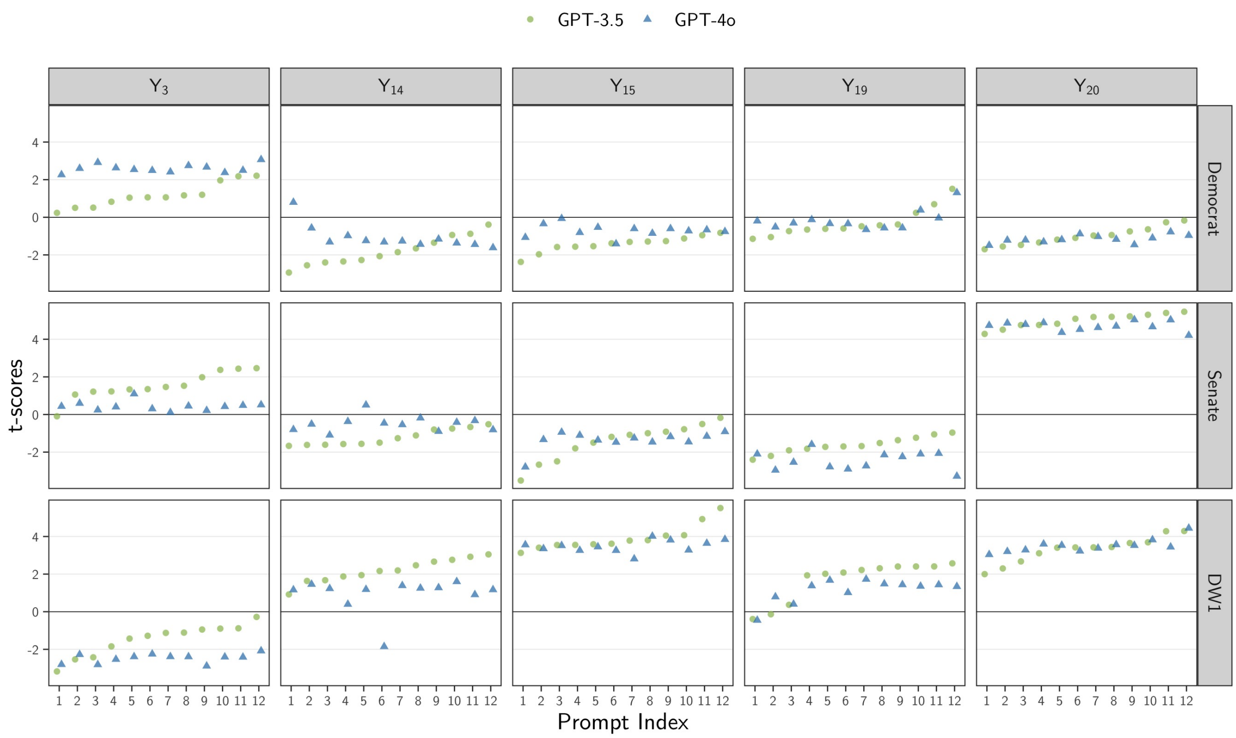
Proxy on the LHS

Figure 1: β_1 vs. Prompt. Proxy on the LHS.



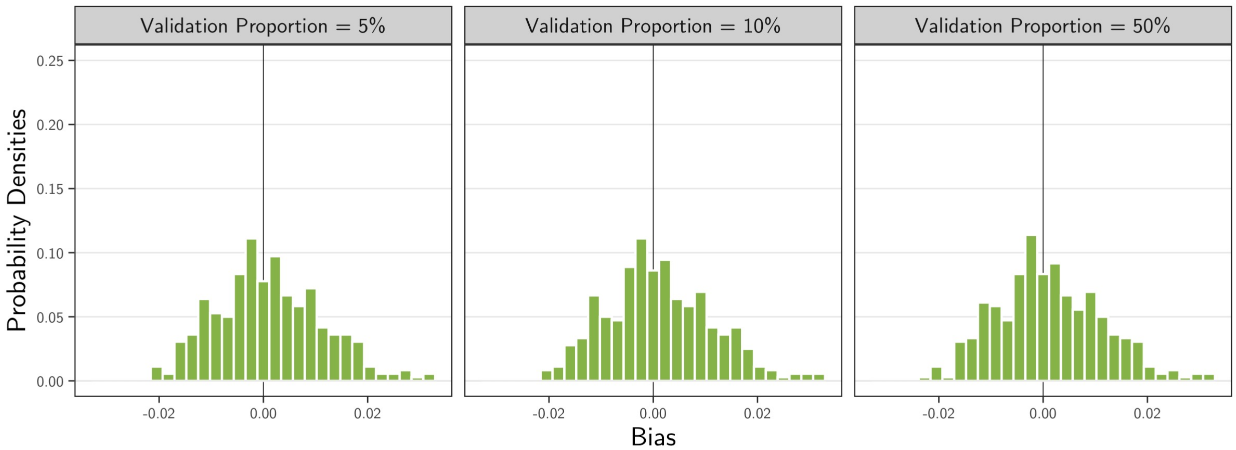
Note: The coefficients were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 2: t-scores of β_1 vs. Prompt. Proxy on the LHS.



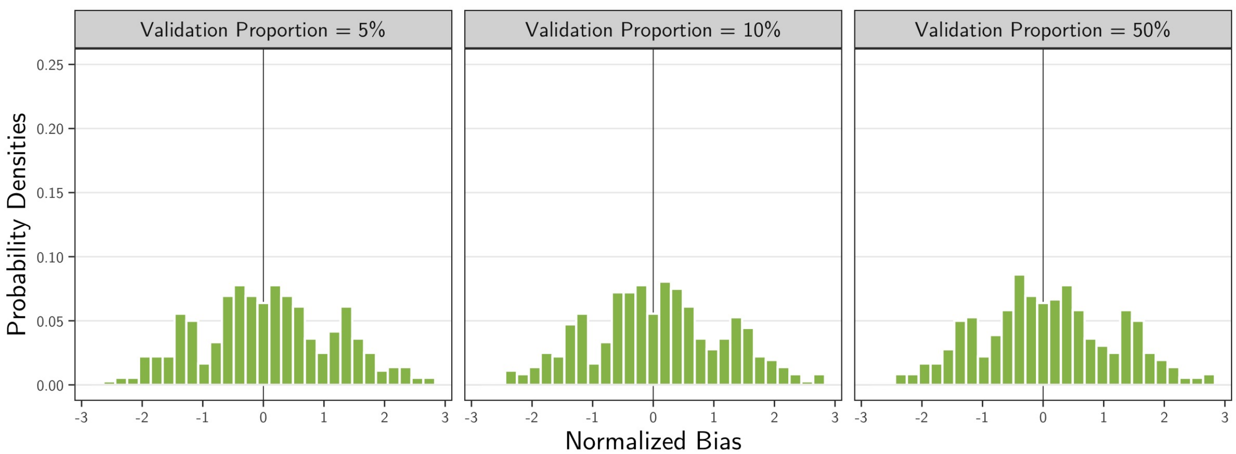
Note: The t-scores were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: Bias Density of β_1^{LLM} by Proportion of Validation Samples. Proxy on the LHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $Y_t = \beta_0 + \beta_1 V$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. A random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$.

Figure 4: Normalized Bias Density of β_1^{LLM} by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 5: Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

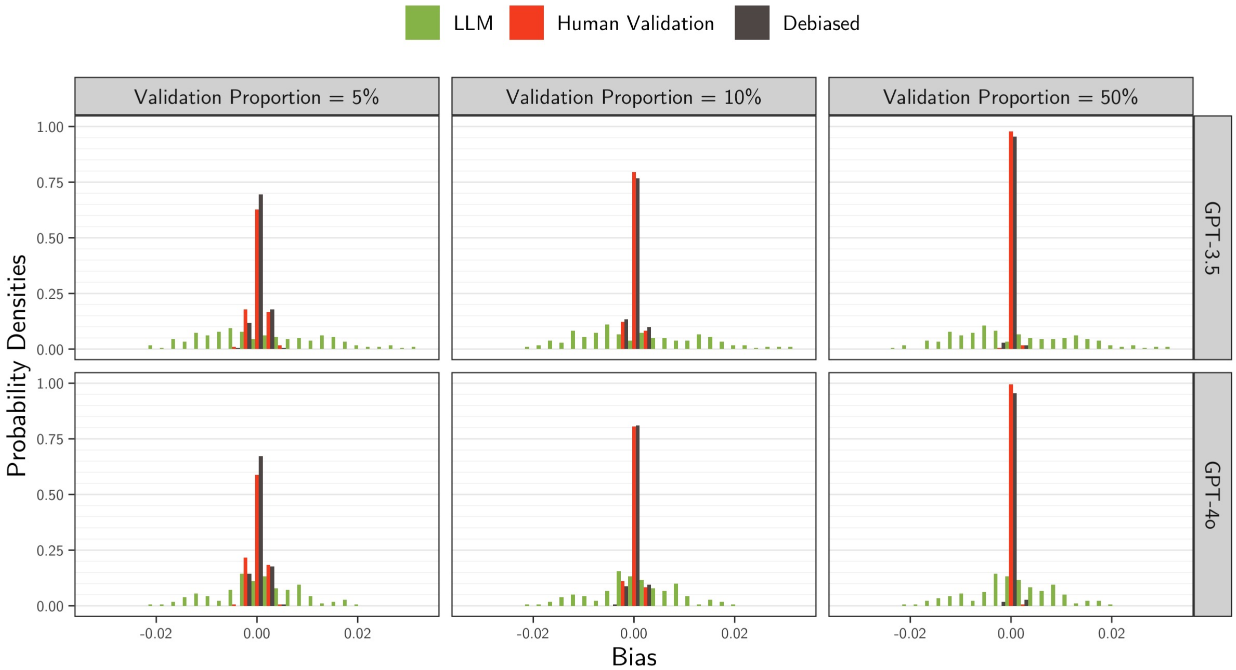


Figure 6: Normalized Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

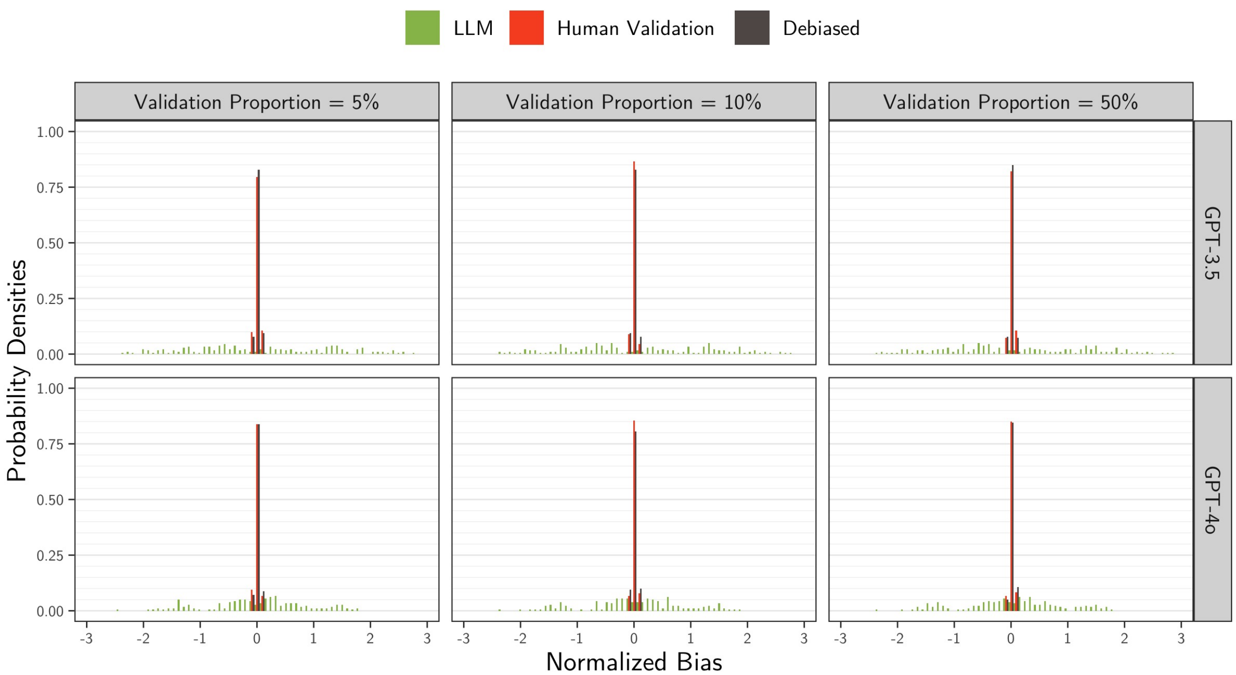


Figure 7: MSE Empirical CDF of β_1 by Proportion of Validation Samples. Proxy on the LHS.

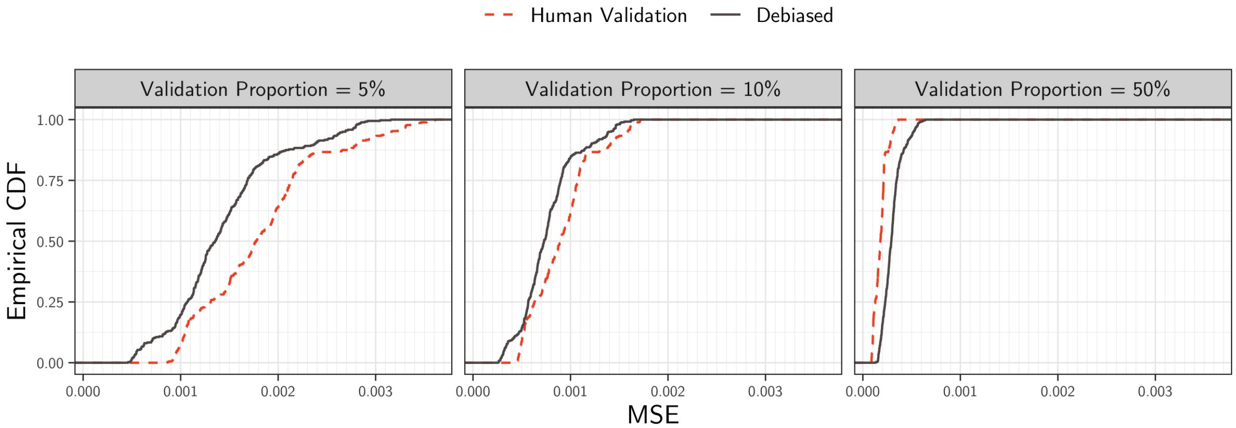


Figure 8: MSE Empirical CDF of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

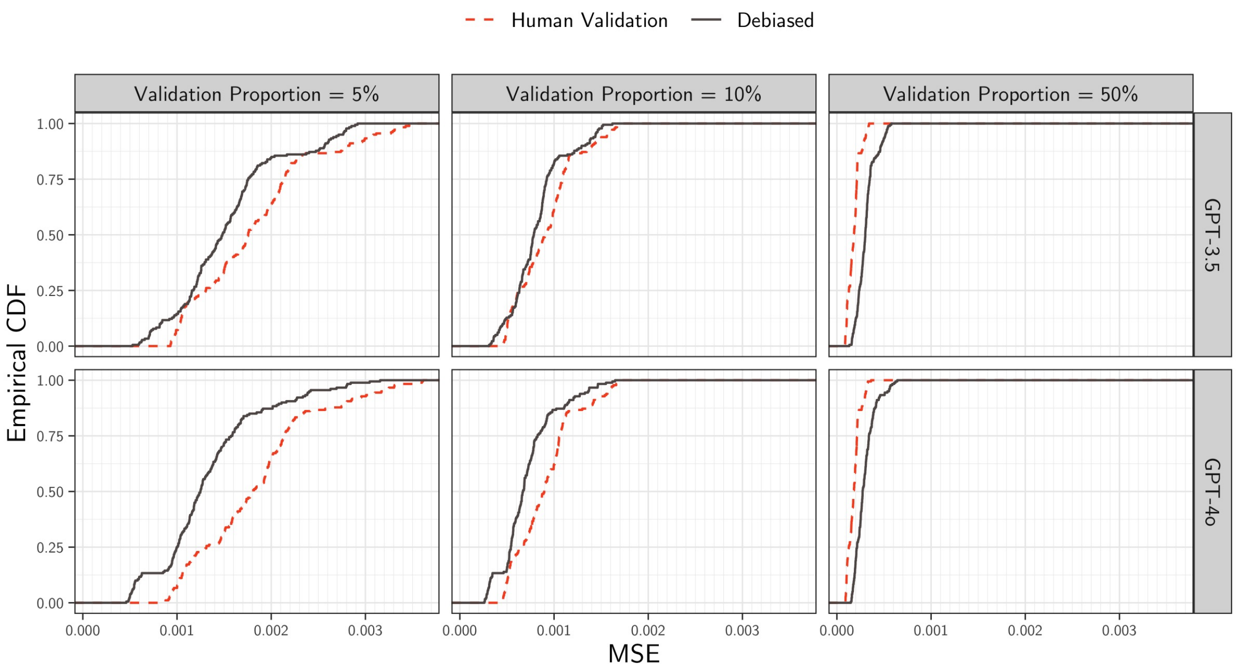


Table 1: Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.010	0.001	-0.015	0.018
5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
5%	Debiased	0.000	0.001	0.000	-0.002	0.002
10%	LLM	0.001	0.010	0.000	-0.015	0.018
10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
10%	Debiased	0.000	0.001	0.000	-0.002	0.001
25%	LLM	0.001	0.010	0.000	-0.014	0.018
25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
25%	Debiased	0.000	0.001	0.000	-0.001	0.001
50%	LLM	0.001	0.010	0.000	-0.014	0.019
50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 2: Normalized Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.071	1.099	0.067	-1.692	1.856
5%	Human Validation	0.000	0.034	0.001	-0.055	0.054
5%	Debiased	0.003	0.033	0.001	-0.055	0.061
10%	LLM	0.069	1.094	0.040	-1.722	1.869
10%	Human Validation	-0.002	0.030	-0.001	-0.055	0.048
10%	Debiased	-0.002	0.033	-0.002	-0.060	0.049
25%	LLM	0.073	1.099	0.023	-1.711	1.854
25%	Human Validation	0.002	0.031	0.001	-0.050	0.051
25%	Debiased	0.001	0.033	0.001	-0.049	0.057
50%	LLM	0.076	1.096	0.047	-1.654	1.875
50%	Human Validation	0.004	0.032	0.003	-0.051	0.056
50%	Debiased	0.000	0.031	-0.001	-0.050	0.053

Table 3: Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	5%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-3.5	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	10%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	25%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	50%	LLM	0.002	0.012	0.000	-0.016	0.022
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	5%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-4o	10%	LLM	0.001	0.008	0.001	-0.014	0.014
GPT-4o	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-4o	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-4o	25%	LLM	0.001	0.008	0.000	-0.013	0.014
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 4: Normalized Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.113	1.284	-0.023	-1.899	2.211
GPT-3.5	5%	Human Validation	0.001	0.034	0.003	-0.052	0.056
GPT-3.5	5%	Debiased	0.003	0.034	0.001	-0.055	0.066
GPT-3.5	10%	LLM	0.109	1.280	-0.005	-1.863	2.179
GPT-3.5	10%	Human Validation	-0.003	0.031	-0.001	-0.064	0.042
GPT-3.5	10%	Debiased	-0.003	0.032	-0.005	-0.053	0.049
GPT-3.5	25%	LLM	0.118	1.285	0.007	-1.837	2.264
GPT-3.5	25%	Human Validation	0.004	0.029	0.002	-0.046	0.050
GPT-3.5	25%	Debiased	0.002	0.031	0.002	-0.044	0.053
GPT-3.5	50%	LLM	0.116	1.282	-0.004	-1.864	2.172
GPT-3.5	50%	Human Validation	0.004	0.033	0.003	-0.052	0.060
GPT-3.5	50%	Debiased	0.000	0.030	0.000	-0.050	0.049
GPT-4o	5%	LLM	0.029	0.877	0.084	-1.411	1.514
GPT-4o	5%	Human Validation	-0.001	0.033	0.000	-0.055	0.051
GPT-4o	5%	Debiased	0.002	0.031	0.001	-0.055	0.054
GPT-4o	10%	LLM	0.028	0.871	0.059	-1.447	1.507
GPT-4o	10%	Human Validation	-0.001	0.030	-0.002	-0.053	0.049
GPT-4o	10%	Debiased	-0.001	0.034	0.000	-0.066	0.050
GPT-4o	25%	LLM	0.027	0.876	0.056	-1.422	1.463
GPT-4o	25%	Human Validation	0.001	0.032	-0.001	-0.050	0.052
GPT-4o	25%	Debiased	0.000	0.035	-0.001	-0.053	0.060
GPT-4o	50%	LLM	0.035	0.874	0.070	-1.441	1.510
GPT-4o	50%	Human Validation	0.004	0.031	0.003	-0.050	0.051
GPT-4o	50%	Debiased	0.000	0.031	-0.001	-0.045	0.059

Table 5: MSE Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0.000	0.000
5%	Human Validation	0.002	0.001	0.002	0.001	0.003
5%	Debiased	0.001	0.001	0.001	0.001	0.003
10%	LLM	0.000	0.000	0.000	0.000	0.000
10%	Human Validation	0.001	0.000	0.001	0.000	0.002
10%	Debiased	0.001	0.000	0.001	0.000	0.001
25%	LLM	0.000	0.000	0.000	0.000	0.000
25%	Human Validation	0.000	0.000	0.000	0.000	0.001
25%	Debiased	0.000	0.000	0.000	0.000	0.001
50%	LLM	0.000	0.000	0.000	0.000	0.000
50%	Human Validation	0.000	0.000	0.000	0.000	0.000
50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 6: MSE Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-3.5	5%	Debiased	0.002	0.001	0.001	0.001	0.003
GPT-3.5	10%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-3.5	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-3.5	50%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-4o	5%	Debiased	0.001	0.001	0.001	0.001	0.002
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-4o	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 7: Coverage Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.811	0.163	0.889	0.461	0.951
5%	Human Validation	0.946	0.008	0.946	0.932	0.959
5%	Debiased	0.928	0.012	0.929	0.904	0.945
10%	LLM	0.810	0.165	0.889	0.460	0.951
10%	Human Validation	0.948	0.007	0.949	0.936	0.959
10%	Debiased	0.940	0.008	0.941	0.926	0.952
25%	LLM	0.809	0.164	0.887	0.465	0.952
25%	Human Validation	0.949	0.007	0.949	0.936	0.960
25%	Debiased	0.946	0.007	0.946	0.934	0.957
50%	LLM	0.810	0.164	0.887	0.463	0.950
50%	Human Validation	0.950	0.007	0.950	0.939	0.961
50%	Debiased	0.948	0.008	0.948	0.935	0.959

Table 8: Coverage Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.761	0.187	0.819	0.381	0.945
GPT-3.5	5%	Human Validation	0.946	0.008	0.948	0.933	0.961
GPT-3.5	5%	Debiased	0.929	0.011	0.931	0.910	0.945
GPT-3.5	10%	LLM	0.759	0.189	0.812	0.383	0.949
GPT-3.5	10%	Human Validation	0.947	0.007	0.947	0.936	0.959
GPT-3.5	10%	Debiased	0.940	0.008	0.941	0.927	0.952
GPT-3.5	25%	LLM	0.760	0.188	0.815	0.362	0.945
GPT-3.5	25%	Human Validation	0.949	0.007	0.949	0.936	0.958
GPT-3.5	25%	Debiased	0.946	0.007	0.946	0.934	0.957
GPT-3.5	50%	LLM	0.760	0.190	0.815	0.369	0.950
GPT-3.5	50%	Human Validation	0.951	0.007	0.950	0.939	0.962
GPT-3.5	50%	Debiased	0.947	0.008	0.948	0.934	0.959
GPT-4o	5%	LLM	0.861	0.116	0.921	0.637	0.954
GPT-4o	5%	Human Validation	0.945	0.008	0.946	0.930	0.957
GPT-4o	5%	Debiased	0.926	0.014	0.927	0.902	0.945
GPT-4o	10%	LLM	0.860	0.117	0.919	0.642	0.952
GPT-4o	10%	Human Validation	0.949	0.007	0.949	0.936	0.959
GPT-4o	10%	Debiased	0.940	0.008	0.941	0.926	0.953
GPT-4o	25%	LLM	0.859	0.117	0.921	0.625	0.954
GPT-4o	25%	Human Validation	0.949	0.007	0.949	0.936	0.960
GPT-4o	25%	Debiased	0.946	0.007	0.946	0.934	0.958
GPT-4o	50%	LLM	0.860	0.115	0.919	0.635	0.950
GPT-4o	50%	Human Validation	0.950	0.007	0.949	0.939	0.961
GPT-4o	50%	Debiased	0.948	0.007	0.948	0.935	0.959